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(54) **ELECTROSTATIC CHUCKS WITH HYBRID PUCKS TO IMPROVE THERMAL PERFORMANCE AND LEAKAGE CURRENT STABILITY**

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H01L 21/67 (2006.01)
H01L 21/687 (2006.01)

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CPC **H01L 21/6833** (2013.01); **H01L 21/67109** (2013.01); **H01L 21/68785** (2013.01)

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See application file for complete search history.

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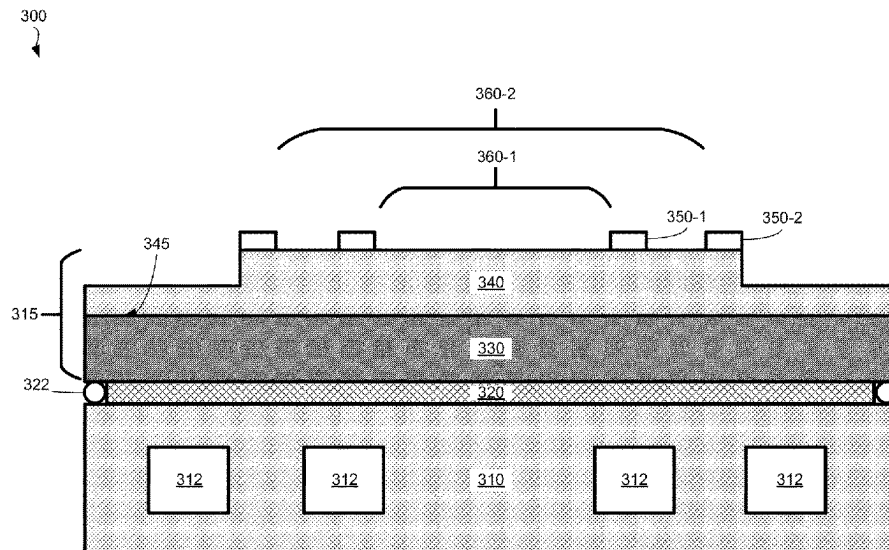
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(57) **ABSTRACT**

A device includes a hybrid puck corresponding to an electrostatic chuck. The hybrid puck includes a backing region and a chucking region disposed on the backing region. The backing region includes a first dielectric material to improve thermal performance of the hybrid puck. The chucking region includes a second dielectric material different from the first dielectric material to improve leakage current stability.

20 Claims, 9 Drawing Sheets



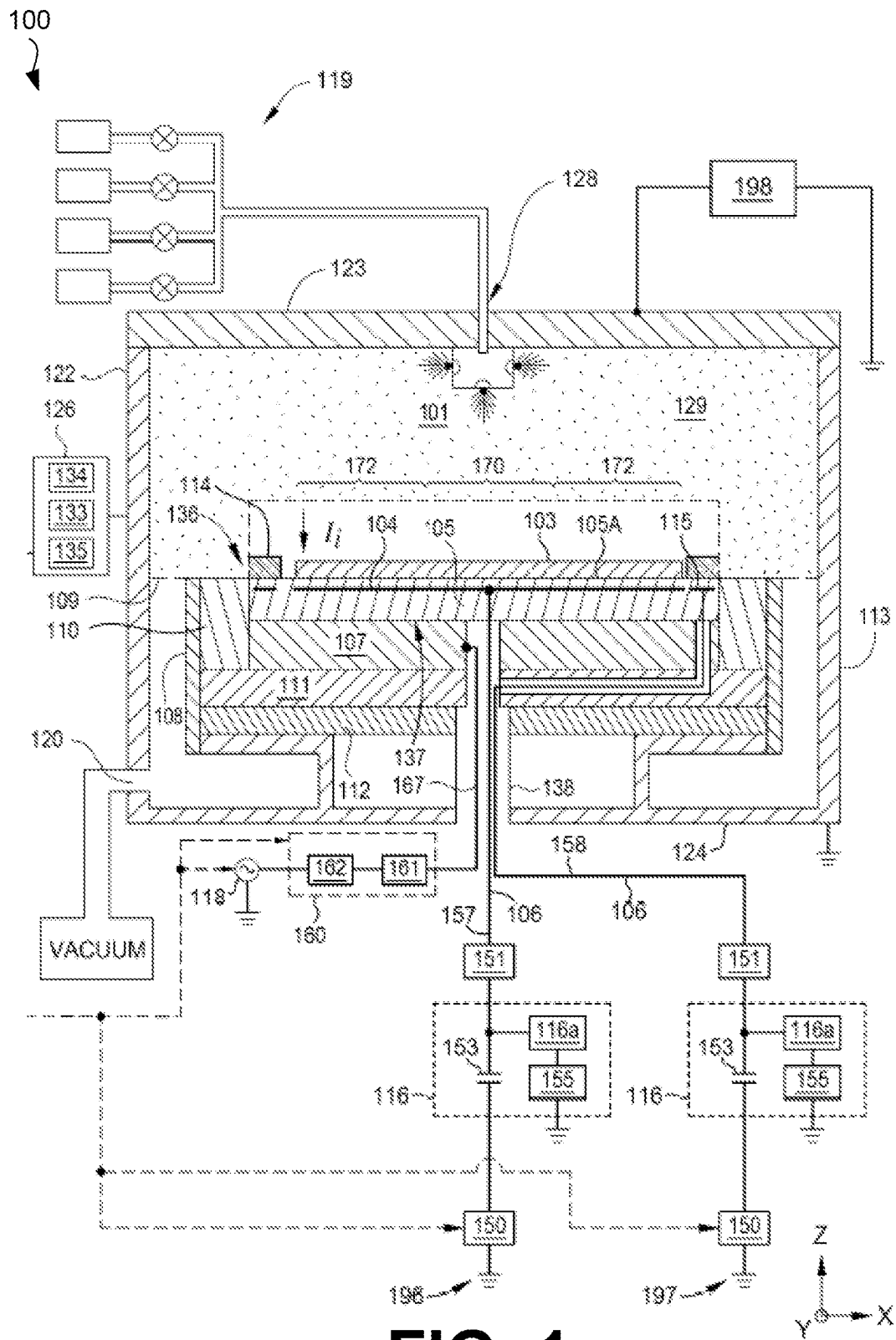


FIG. 1

200

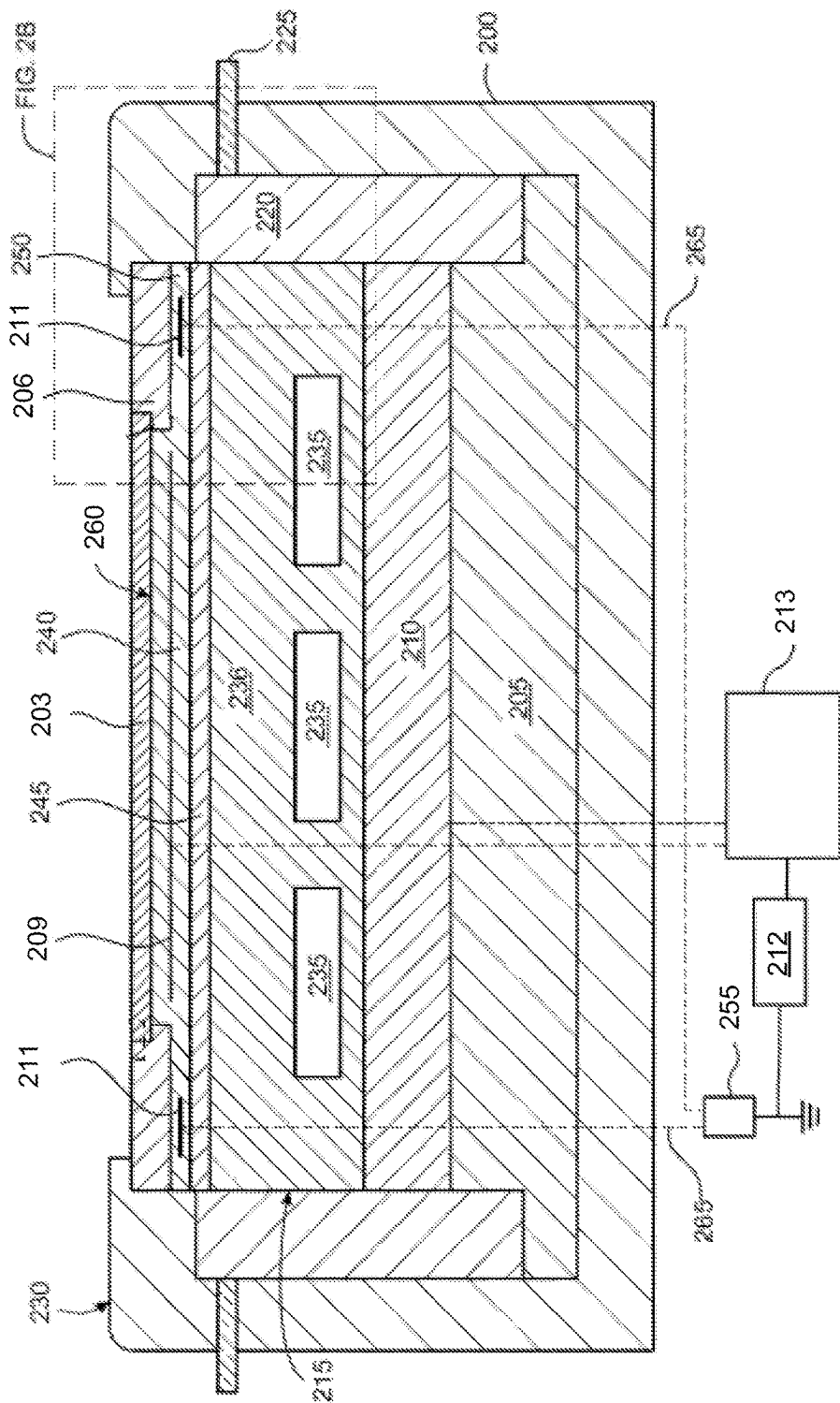


FIG. 2A

200

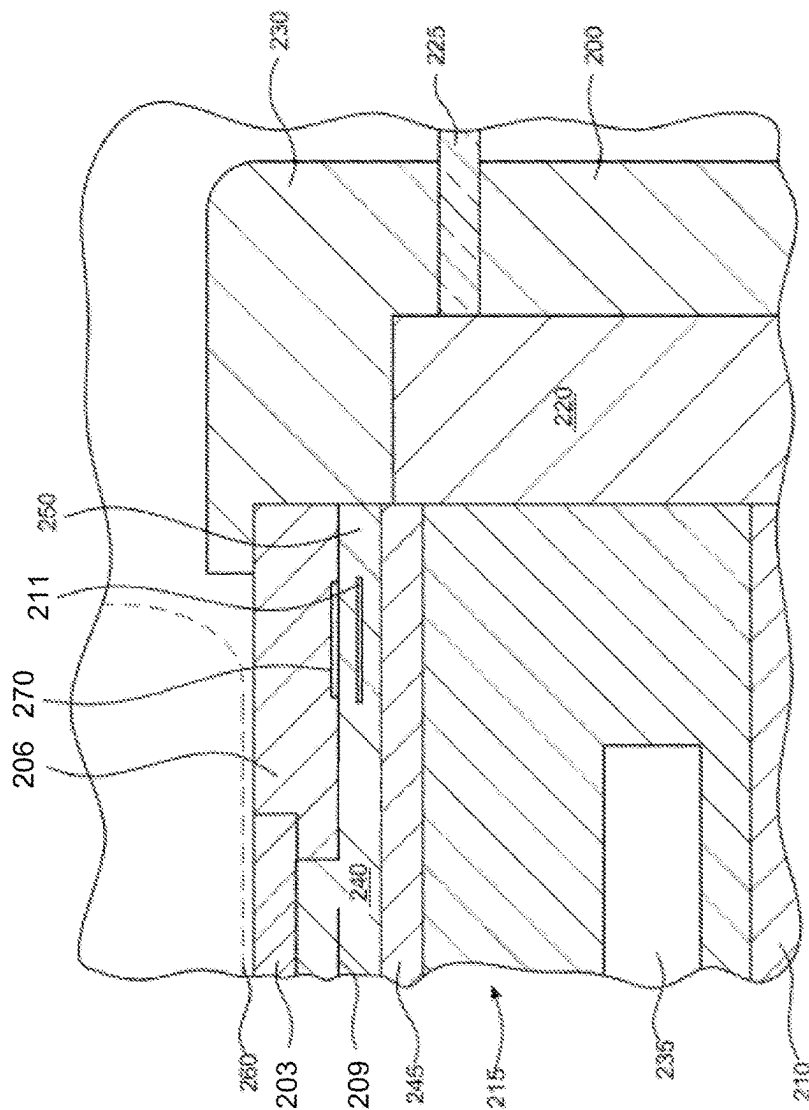


FIG. 2B

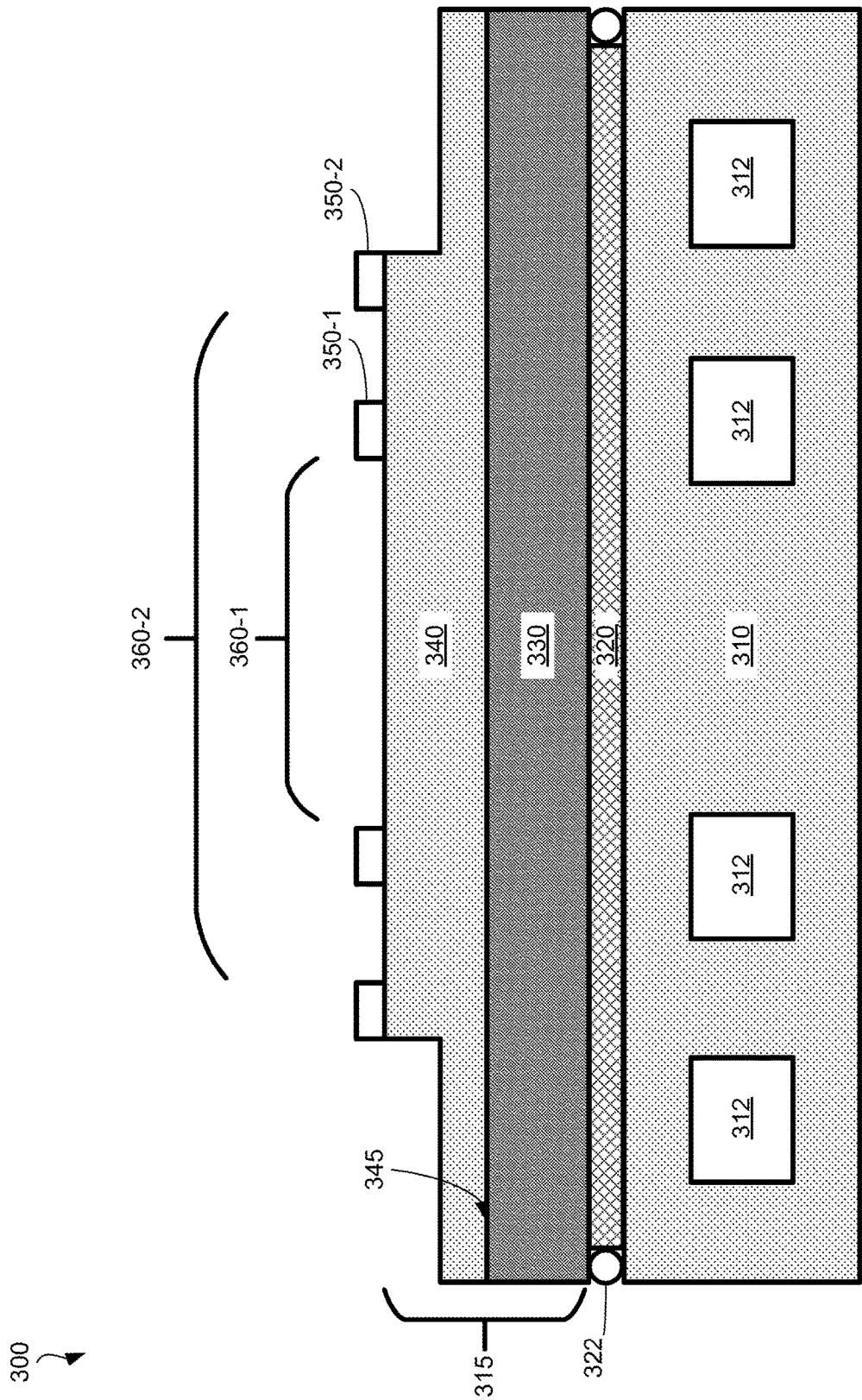


FIG. 3A

315

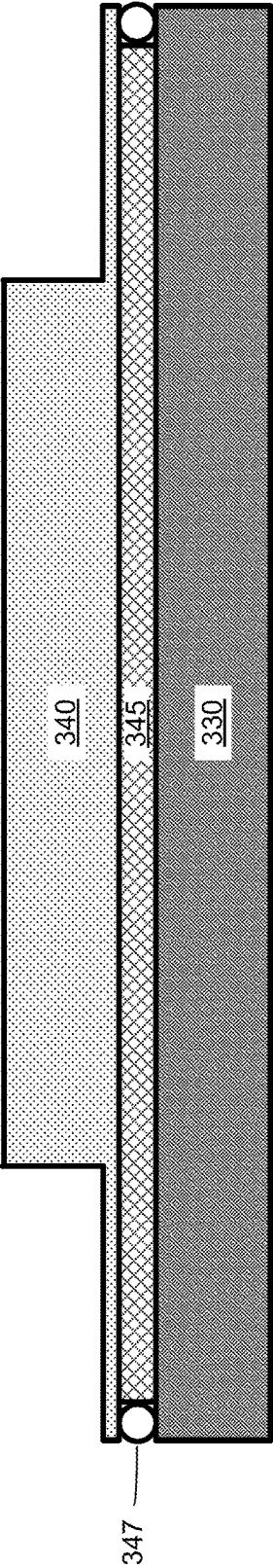


FIG. 3B

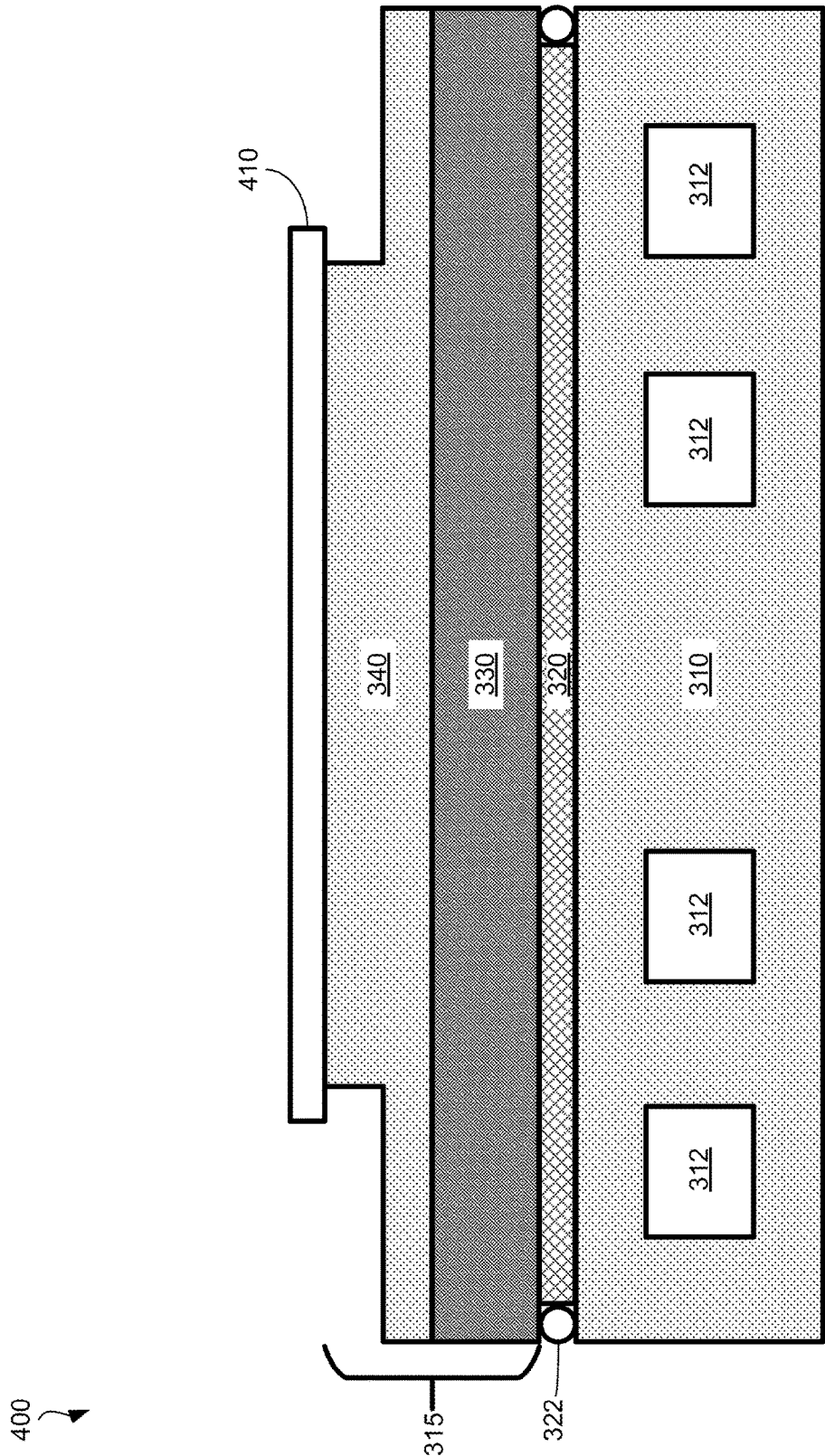


FIG. 4

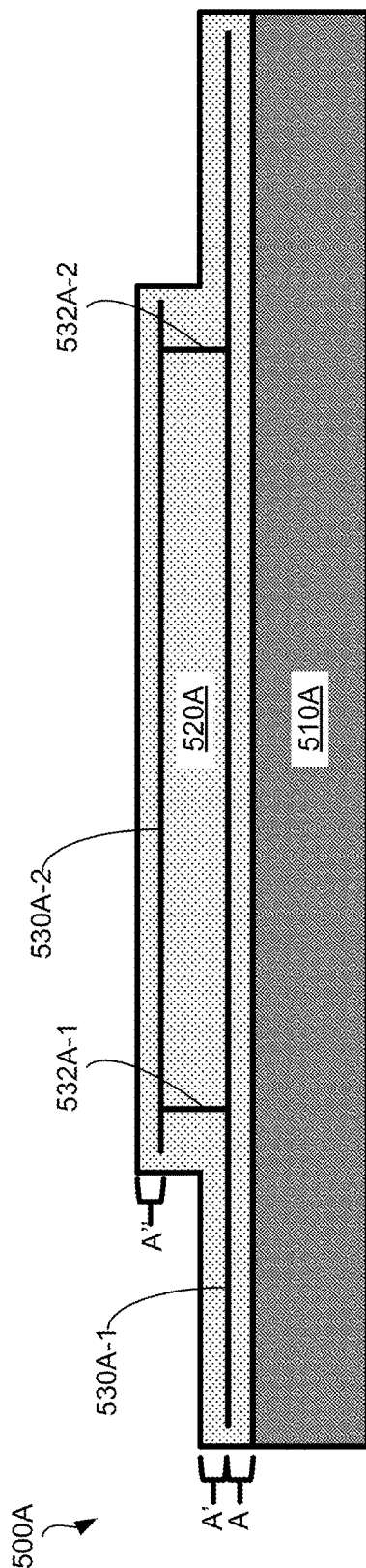


FIG. 5A

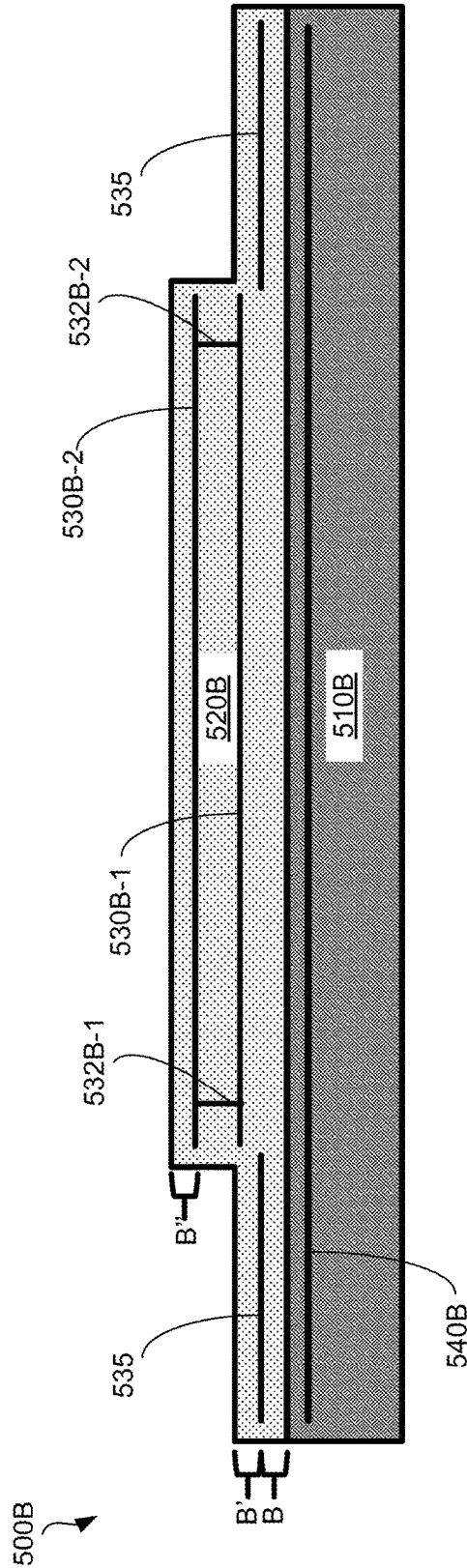


FIG. 5B

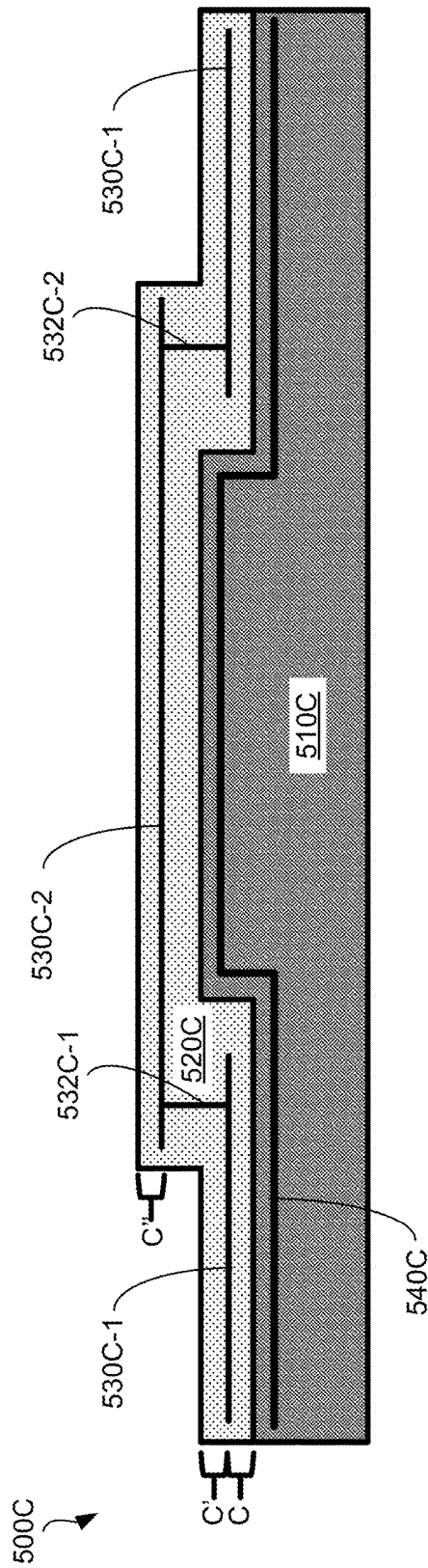
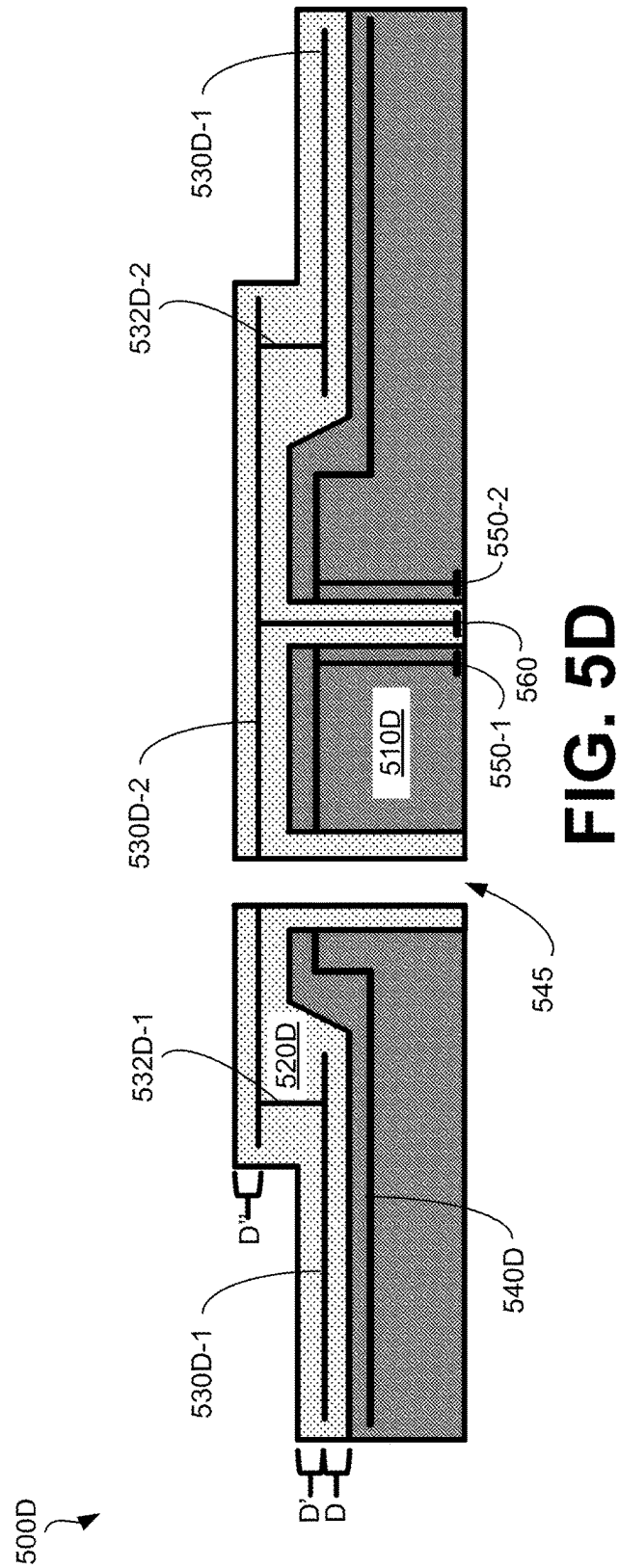
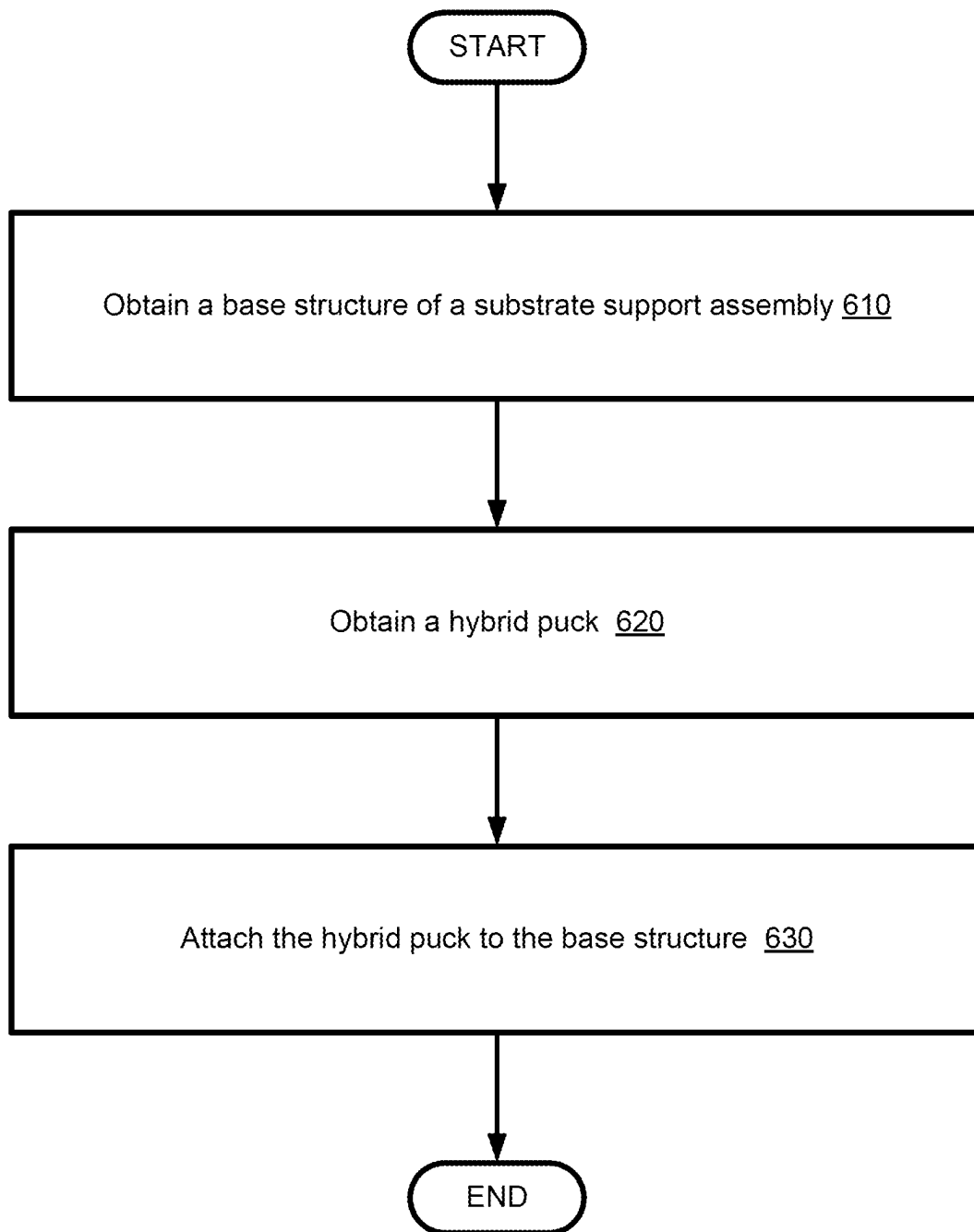


FIG. 5C



600

**FIG. 6**

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ELECTROSTATIC CHUCKS WITH HYBRID PUCKS TO IMPROVE THERMAL PERFORMANCE AND LEAKAGE CURRENT STABILITY

TECHNICAL FIELD

Embodiments of the present invention relate, in general, to substrate processing, and in particular, to electrostatic chucks (ESCs) with hybrid pucks to implement electrostatic chucks with improved thermal performance and stability.

BACKGROUND

An electronic device manufacturing apparatus can include multiple chambers, such as processing chambers and load lock chambers. Such an electronic device manufacturing apparatus can employ a robot apparatus in the transfer chamber that is configured to transport substrates between the multiple chambers. In some instances, multiple substrates are transferred together. Processing chambers may be used in an electronic device manufacturing apparatus to perform one or more processes on substrates, such as deposition processes, etch processes and/or lithography processes. An electrostatic chuck (ESC) is a device that can generate electrostatic force to securely hold a substrate (e.g., wafer) in place against a puck without requiring physical force during one or more processes, such as during deposition, etching and/or lithography processes. Utilizing electrostatic force, without requiring physical force, can reduce the risk of damage to the substrate during processing and can create a more stable and/or uniform hold as compared to other chucks (e.g., mechanical chucks).

SUMMARY

In some embodiments, a device is provided. The device includes a hybrid puck corresponding to an electrostatic chuck. The hybrid puck includes a backing region and a chucking region disposed on the backing region. The backing region includes a first dielectric material to improve thermal performance of the hybrid puck. The chucking region includes a second dielectric material different from the first dielectric material to improve leakage current stability.

In some embodiments, a substrate support assembly is provided. The substrate support assembly includes a base structure including a cooling plate having a plurality of cooling channels, and a hybrid puck corresponding to an electrostatic chuck and disposed on the base structure. The hybrid puck includes a backing region disposed on the base structure and a chucking region disposed on the backing region. The backing region includes a first dielectric material to improve thermal performance of the hybrid puck. The chucking region includes a second dielectric material different from the first dielectric material to improve leakage current stability.

In some embodiments, a method is provided. The method includes obtaining a base structure including a cooling plate having a plurality of cooling channels, and attaching, to the base structure, a hybrid puck including a backing region disposed on the base structure and a chucking region disposed on the backing region. The backing region includes a first dielectric material to improve thermal performance of the hybrid puck. The chucking region includes a second dielectric material different from the first dielectric material to improve leakage current stability.

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Numerous other aspects and features are provided in accordance with these and other embodiments of the disclosure. Other features and aspects of embodiments of the disclosure will become more fully apparent from the following detailed description, the claims, and the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments described herein are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings in which like references indicate similar elements. It should be noted that different references to “an” or “one” embodiment in this disclosure are not necessarily to the same embodiment, and such references mean at least one.

FIG. 1 is a diagram of a cross-sectional view of a processing chamber, in accordance with some embodiments.

FIGS. 2A-2B are diagrams of schematic side views of portions of a substrate support assembly, in accordance with some embodiments.

FIGS. 3A, 3B and 4 are diagrams of cross-sectional views of example substrate support assemblies including hybrid pucks to improve thermal performance and leakage current stability, in accordance with some embodiments.

FIGS. 5A-5D are diagrams of cross-sectional views of hybrid pucks to improve thermal performance and leakage current stability, in accordance with some embodiments.

FIG. 6 is a flow chart of an example method for fabricating electrostatic chucks (ESCs) including hybrid pucks, in accordance with some embodiments.

DETAILED DESCRIPTION OF EMBODIMENTS

Described herein are embodiments of electrostatic chucks (ESCs) with hybrid pucks to improve thermal performance and leakage current stability. An ESC can include a flat plate, or puck, with a set of electrodes embedded in the surface of the puck. When a voltage is applied to the set of electrodes, an electrostatic field having a strength proportional to the applied voltage is created between the puck and the substrate as well as the distance between the surfaces of the puck and the substrate. Thus, when the applied voltage is sufficiently high, the electrostatic field can have sufficient strength to generate an electrostatic force that securely holds the substrate in place on the puck. ESCs can be designed to accommodate various different substrates sizes and/or shapes. For example, an ESC can have ring-shaped electrodes embedded within the puck to hold circular substrates. As another example, an ESC can have a grid pattern of electrodes embedded within the puck to hold square or rectangular substrates. The puck can be disposed on a cooling plate to enable substrate cooling.

Some pucks are formed from a single, monolithic dielectric material, such as a ceramic material. Although dielectric materials serve as electrical insulators, different dielectric materials can have different auxiliary properties. For example, some dielectric materials can have high thermal conductivity for improved substrate cooling using the cooling plate. As another example, some dielectric materials can offer leakage current stability. Leakage currents are unwanted electrical currents that flow through a device, which can be caused by defects within the material of the device that enable electron flow. Accordingly, some ESCs can sacrifice thermal conductivity for improved leakage current protection (or vice versa).

To address at least the above-noted drawbacks, embodiments described herein provide for ESCs with hybrid pucks to improve thermal performance and leakage current stability. A hybrid puck described herein can include a chucking region having a chucking surface, and a backing region underneath the chucking region. The backing region can be formed from a first dielectric material and the chucking region can be formed from a second dielectric material different from the first dielectric material. More specifically, the first dielectric material can be selected to improve thermal performance at the interface between the puck and the cooling plate. The second dielectric material can be selected to improve stability and leakage current consistency for the chucking surface. In some embodiments, at least one of the first dielectric material or the second dielectric material is a ceramic material. In some embodiments, first dielectric material includes aluminum nitride (AlN). In some embodiments, the second dielectric material includes aluminum oxide or alumina (Al₂O₃). In some embodiments, the chucking region and the backing region are bonded together with a bond. The chucking region and the backing region can be bonded together with any suitable type of bond. Examples of bonds include diffusion bonds, metal bonds, etc. Additionally, an ESC described herein can include a set of electrodes can be embedded within the puck to secure a substrate to the surface of the puck. An ESC can further include a cooling plate disposed underneath the puck, where the cooling plate includes a set of cooling channels to cool the substrate. For example, the puck can be bonded to the cooling plate using a bonding layer. The puck can include an outer cooling zone defined by an outer seal band and an inner cooling zone defined by an inner seal band. The hybrid puck can help reduce the temperature drop across the dielectric material to achieve a lower substrate temperature for a given cooling plate and a given coolant fluid condition. Further details regarding ESCs with hybrid pucks to improve thermal performance and leakage current stability are described herein below with reference to FIGS. 1-6.

FIG. 1 is a cross-sectional view of a processing chamber 100, in accordance with some embodiments. The plasma processing chamber 100 is configured to practice one or more of the biasing schemes proposed herein, according to one or more embodiments. In one embodiment, the processing chamber 100 is a plasma processing chamber, such as a reactive ion etch (RIE) plasma chamber. In some other embodiments, the processing chamber 100 is a plasma-enhanced deposition chamber, for example a plasma-enhanced chemical vapor deposition (PECVD) chamber, a plasma enhanced physical vapor deposition (PEPVD) chamber, or a plasma-enhanced atomic layer deposition (PEALD) chamber. In some embodiments, the processing chamber 100 is a plasma treatment chamber or a plasma based ion implant chamber (e.g., a plasma doping (PLAD) chamber). In some embodiments, the plasma source is a capacitively coupled plasma (CCP) source, which includes an electrode (e.g., chamber lid 123) disposed in the processing volume facing the substrate support assembly 136. In some configurations, an opposing electrode, such as the chamber lid 123, which is positioned opposite to the substrate support assembly 136, is electrically coupled to ground. However, as illustrated in FIG. 1, the opposing electrode can alternately be electrically coupled to a source 198 that is coupled to ground, such as a source that includes an RF generator or a PV generator. In some embodiments, the processing chamber 100 may additionally or alternatively, include an inductively coupled plasma (ICP) source electrically coupled to a radio frequency (RF) power supply.

The processing chamber 100 also includes a chamber body 113 that includes the chamber lid 123, one or more sidewalls 122, and a chamber base 124, which define a processing volume 129. The one or more sidewalls 122 and chamber base 124 generally include materials that are sized and shaped to form the structural support for the elements of the plasma processing chamber 100, and are configured to withstand the pressures and added energy applied to them while a plasma 101 is generated within a vacuum environment maintained in the processing volume 129 of the plasma processing chamber 100 during processing. In one example, the one or more sidewalls 122 and chamber base 124 are formed from a metal, such as aluminum, an aluminum alloy, a stainless steel, etc. A gas inlet 128 disposed through the chamber lid 123 is used to provide one or more processing gases to the processing volume 129 from a processing gas source 119 that is in fluid communication therewith. A substrate 103 is loaded into, and removed from, the processing volume 129 through an opening (not shown) in one of the one or more sidewalls 122, which is sealed with a slit valve (not shown) during plasma processing of the substrate 103. Herein, the substrate 103 is transferred to and from a substrate receiving surface 105A of an ESC substrate support 105 using a lift pin system (not shown).

In some embodiments, an RF generator assembly 160 is configured to deliver RF power to the support base 107 disposed proximate to the ESC substrate support 105, and within the substrate support assembly 136. The RF power delivered to the support base 107 is configured to ignite and maintain a processing plasma 101 formed by use of process gases disposed within the processing volume 129. In some embodiments, the support base 107 is an RF electrode that is electrically coupled to an RF generator 118 via an RF matching circuit 162 and a first filter assembly 161, which are both disposed within the RF generator assembly 160. In some embodiments, the plasma generator assembly 160 and RF generator 118 are used to ignite and maintain a processing plasma 101 using the processing gases disposed in the processing volume 129 and fields generated by the RF power provided to the support base 107 by the RF generator 118. The processing volume 129 is fluidly coupled to one or more dedicated vacuum pumps, through a vacuum outlet 120, which maintain the processing volume 129 at sub-atmospheric pressure conditions and evacuate processing and/or other gases, therefrom. A substrate support assembly 136, disposed in the processing volume 129, is disposed on a support shaft 138 that is grounded and extends through the chamber base 124. However, in some embodiments, the RF generator assembly 160 is configured to deliver RF power to the biasing electrode 104 disposed in the substrate support 105 versus the support base 107.

The substrate support assembly 136, as briefly discussed above, generally includes a substrate support 105 (e.g., ESC substrate support) and support base 107. In some embodiments, the substrate support assembly 136 can additionally include an insulator plate 111 and a ground plate 112. The substrate support 105 is thermally coupled to and disposed on the support base 107. In some embodiments, the support base 107 is configured to regulate the temperature of the substrate support 105, and the substrate 103 disposed on the substrate support 105, during substrate processing. In some embodiments, the support base 107 includes a cooling plate including one or more cooling channels (not shown in FIG. 1) disposed therein that are fluidly coupled to, and in fluid communication with, a coolant source (not shown), such as a refrigerant source or water source having a relatively high electrical resistance. In some embodiments, the substrate

support **105** includes a heater electrode (not shown in FIG. 1), such as a resistive heating element, embedded in the dielectric material thereof. The support base **107** can be formed of a corrosion resistant thermally conductive material, such as a corrosion resistant metal, for example aluminum, an aluminum alloy, a stainless steel, etc. The support base **107** can be coupled to the substrate support **105** with an adhesive or by mechanical means (e.g., fasteners).

The support base **107** is electrically isolated from the chamber base **124** by the insulator plate **111**, and the ground plate **112** is interposed between the insulator plate **111** and the chamber base **124**. In some embodiments, the processing chamber **100** further includes a quartz pipe **110**, or collar, that at least partially circumscribes portions of the substrate support assembly **136** to prevent corrosion of the ESC substrate support **105** and, or, the support base **107** from contact with corrosive process gases or plasma, cleaning gases or plasma, or byproducts thereof. Typically, the quartz pipe **110**, the insulator plate **111**, and the ground plate **112** are circumscribed by a liner **108**. Herein, a plasma screen **109** approximately coplanar with the substrate receiving surface of the ESC substrate support **105** prevents plasma from forming in a volume between the liner **108** and the one or more sidewalls **122**.

The substrate support **105** can be formed from a dielectric material, such as a bulk sintered ceramic material, such as a corrosion resistant metal oxide or metal nitride material, for example aluminum oxide (Al_2O_3), aluminum nitride (AlN), titanium oxide

(TiO), titanium nitride (TiN), yttrium oxide (Y_2O_3), or combinations thereof. In embodiments herein, the substrate support **105** further includes a biasing electrode **104** embedded in the dielectric material thereof. In one configuration, the biasing electrode **104** is a chucking pole used to secure (chuck) the substrate **103** to a substrate receiving surface **105A** of the substrate support **105**, also referred to herein as an ESC substrate support, and to bias the substrate **103** with respect to the processing plasma **101** using one or more of the pulsed-voltage biasing schemes described herein.

The biasing electrode **104** can be formed from one or more electrically conductive parts, such as one or more metal meshes, foils, plates, or combinations thereof. In some embodiments, the biasing electrode **104** is electrically coupled to a bias compensation module **116**, which provides a chucking voltage thereto, such as static DC voltage between about -5000 V and about 5000 V, using an electrical conductor, such as the coaxial transmission line **106** (e.g., a coaxial cable). The high voltage module **116** includes bias compensation circuit elements **116A**, a DC power supply **155**, and a blocking capacitor **153**. A bias compensation module blocking capacitor, which is also referred to herein as the blocking capacitor **153**, is disposed between the output of a pulsed-voltage waveform generator (PVWG) **150** and the biasing electrode **104**.

The biasing electrode **104** is spaced apart from the substrate receiving surface **105A** of the substrate support **105**, and thus from the substrate **103**, by a layer of dielectric material of the substrate support **105**. Depending on the type of electrostatic chucking method utilized within the substrate support **105** to retain a substrate **103** during processing, such as a coulombic ESC or a Johnsen-Rahbek ESC, the effective circuit elements used to model the electrical coupling of the biasing electrode **104** to the plasma **101** will vary. Herein, the biasing electrode **104** is electrically coupled to the output of the pulsed-voltage waveform gen-

erator (PVWG) **150** using the external conductor, such as the transmission line **106**, which is disposed within the support shaft **138**.

The substrate support assembly **136** can further include an edge control electrode **115** formed from a conductive material that is positioned below the edge ring **114** (e.g., process kit) and surrounds the biasing electrode **104**. For example, edge control electrode can adjust on-wafer etch feature tilting at the edge. When the edge control electrode **115** is biased, due to its position relative to the substrate **103**, it can affect or alter a portion of the generated plasma **101** that is at or outside of the edge of the substrate **103**. The edge control electrode **115** can be biased by use of a PVWG **150** that is different from the PVWG **150** that is used to bias the biasing electrode **104**. In some embodiments, a first PV waveform generator **150** of a first PV source assembly **196** is configured to bias the biasing electrode **104**, and a second PV waveform generator **150** of a second PV source assembly **197** is configured to bias the edge control electrode **115**. In some embodiments, the edge control electrode **115** is positioned within a region of the substrate support **105**, as shown in FIG. 1. In general, if processing chambers **100** is configured to process circular substrates, the edge control electrode **115** can be annular in shape (e.g., a ring electrode) and can be configured to surround at least a portion of the biasing electrode **104**. In some embodiments, as illustrated in FIG. 1, the edge control electrode **115** includes a conductive mesh, foil or plate that is disposed a similar distance (i.e., Z-direction) from the surface **105A** of the substrate support **105** as the biasing electrode **104**. In some other embodiments, the edge control electrode **115** includes a conductive mesh, foil or plate that is positioned on or within a region of the dielectric pipe **110** (e.g., AlN, or Al_2O_3), which surrounds at least a portion of the biasing electrode **104** and/or the substrate support **105**. Alternately, in some embodiments, the edge control electrode **115** is positioned within or is coupled to the edge ring **114**, which is disposed adjacent to the substrate support **105**. In this configuration, the edge ring **114** is formed from a semiconductor or dielectric material (e.g., AlN or Al_2O_3).

The plasma processing chamber **100** further includes a controller **126**. The controller **126** can include a central processing unit (CPU) **133**, a memory **134**, and support circuits **135**. The controller **126** is used to control the process sequence used to process the substrate **103** including the substrate biasing methods described herein. The CPU **133** is a general-purpose computer processor configured for use in an industrial setting for controlling processing chamber and sub-processors related thereto. The memory **134**, which is generally non-volatile memory, may include random access memory, read only memory, floppy or hard disk drive, or other suitable forms of digital storage, local or remote. The support circuits **135** can be coupled to the CPU **133** and comprise cache, clock circuits, input/output subsystems, power supplies, and the like, and combinations thereof. Software instructions (program) and data can be coded and stored within the memory **134** for instructing a processor within the CPU **133**. A software program (or computer instructions) readable by CPU **133** in the controller **126** determines which tasks are performable by the components in the plasma processing chamber **100**. Preferably, the program, which is readable by CPU **133** in the controller **126**, includes code, which when executed by the processor (CPU **133**), perform tasks relating to the monitoring and execution of the electrode biasing scheme described herein. The program will include instructions that are used to control the various hardware and electrical components

within the plasma processing chamber **100** to perform the various process tasks and various process sequences used to implement the electrode biasing scheme described herein.

During processing, the PV generators within the PV waveform generators **150** of the first PV source assembly **196** and the second PV source assembly **197** establishes a pulsed voltage waveform on the load disposed with the plasma processing chamber **100**. The overall control of the delivery of the PV waveform from each of the PV waveform generators **150** is controlled by use of signals provided from the controller **126**. Each PV waveform generator **150** will include a PV generator (e.g., DC power supply) and one or more electrical components, such as high repetition rate switches, capacitors (not shown), inductors (not shown), fly back diodes (not shown), power transistors (not shown) and/or resistors (not shown), that are configured to provide a PV waveform.

The PV transmission line **157** electrically connects the output of the PV waveform generator **150** to the second filter assembly **151**. The PV transmission line **158** of the second PV source assembly **197** couples a PV waveform generator **150** to the edge control electrode **115**, will include the same or similar components. The electrical conductor(s) within the various parts of the PV transmission line **157**, **158** may include elements such as: (a) a coaxial transmission line (e.g., coaxial line **106**), which may include a flexible coaxial cable that is connected in series with a rigid coaxial transmission line, (b) an insulated high-voltage corona-resistant hookup wire, (c) a bare wire, (d) a metal rod, (e) an electrical connector, or (f) any combination thereof.

FIGS. 2A and 2B illustrate enlarged schematic side views of portions of a substrate support assembly **204**, according to some embodiments. For example, substrate support assembly **204** can be similar to substrate support assembly **204** of FIG. 1. The substrate support assembly **204** includes a ground plate **200** surrounding an insulating layer **205**, a facilities plate **210**, and an ESC **215** assembled in a vertical stack. A quartz pipe ring **220** circumscribes the facilities plate **210** and the ESC **215** to insulate the ESC **215** from the ground plate **200**. The ESC **215** includes one or more chucking electrodes (e.g., the first electrode **209**) embedded therein for chucking the substrate **203** to a support surface of the ESC **215**. A plasma shield **225** is disposed on an upper surface of the ground plate **200** to facilitate plasma containment in a process chamber. A quartz ring **230** is positioned on an upper surface of the plasma shield **225**.

The facilities plate **210** can be made of a conductive material, such as aluminum, or other suitable conductive material, and is positioned between a lower portion of the ground plate **200** and the ESC **215**. The facilities plate **210** is configured to route fluid and/or gas from an input location (e.g., at a bottom thereof not shown) to an output location (e.g., at a top thereof, not shown). The ESC **215** includes one or more channels **235** formed in a first material **236** through which a fluid is provided to facilitate temperature control of the substrate support assembly **204**. The first material **236** is a metallic material, such as aluminum. The ESC **215** includes the first electrode **209** embedded in a second material **240**. The second material **240** is a dielectric material, such as a ceramic material, such as alumina or aluminum nitride. A heater **245** is disposed adjacent to or in the ESC **215** to facilitate temperature control of the substrate **203**. The heater **245** may be, for example, a resistive heater having a plurality of resistive heating elements embedded therein.

An edge control electrode **211** (e.g., ring electrode) is embedded in an edge region **250** of the ESC **215**. For

example, in the illustrated embodiment, the edge control electrode **211** is embedded in the edge region **250** of the second material **240**. In some embodiments, the edge control electrode **211** is a ring electrode.

The edge control electrode **211** may be positioned about 0.3 millimeters to about 1 millimeter from the upper surface of the edge region **250**, such as about 0.75 millimeters. The edge control electrode **211** may have a width of about 3 millimeters to about 20 millimeters, such as about 15 millimeters.

The edge control electrode **211** is positioned radially outward of the perimeter of the substrate **203** and beneath the edge ring **206**. In one example, the edge control electrode **211** may have an inner diameter greater than 200 millimeters, or greater than 300 millimeters, or greater than 450 millimeters. The edge control electrode **211** is electrically coupled to ground and/or matching network **212** through the edge tuning circuit **255** which can include one or more capacitors and/or inductors. The edge control electrode **211** may be coupled to the edge tuning circuit **255** through multiple transmission lines **265** (two are shown). For example, the edge control electrode **211** may be coupled to the edge tuning circuit **255** through three transmission lines **265** spaced about the substrate support assembly **204** at even intervals (e.g., 120 degrees).

The edge ring **206** is positioned on (over) the edge region **250** and in contact with the edge region **250** and the second material **240**. In one example, the edge ring **206** may be formed from silicon carbide, graphite coated with silicon carbide, or low resistivity doped silicon. The edge ring **206** circumscribes the substrate **203** and reduces undesired etching or deposition of material at the radially outward edge of the substrate **203**.

Referring to FIG. 2B, during processing, a plasma sheath **260** may form over the surface of the substrate **203** (shown as a dashed line in FIG. 2B). As described above, processing conditions may erode the upper portion of the edge ring **206**, causing undesired processing of the edge of the substrate **203**, such as rounding, sometimes referred to as a "roll-over effect". The undesired processing reduces device yield and affects center-to-edge uniformity. To reduce these undesired effects, conventional approaches frequently replaced the edge ring **206**. However, frequent replacement of the edge ring **206** is expensive both in terms of the cost of a new edge ring and in terms of the significant down time required for the replacement of the edge ring.

The edge control electrode **211** can be coupled to ground and/or a power source **213**, such as an RF generator or PV waveform generator, through either the edge tuning circuit **255** or the matching network **212** to adjust the RF amplitude (and/or phase), and thus the plasma sheath **260**, near the edge ring **206**. In addition, two or more of the power sources **213** and the matching network **212** are configured to be shared during use by the substrate **203** and the edge control electrode **211** disposed beneath the edge ring **206**.

In some embodiments, a plasma sheath **260** above the edge ring **206** that is thicker or thinner than the plasma sheath **260** above the substrate **203** is desired in order to tune one or a combination of the film etching, deposition profile, or feature tilting angle near the substrate edge. Controlling the RF amplitude and/or RF phase at the edge ring **206** relative to the RF amplitude and/or RF phase at the substrate **203** allows such process edge profile tuning.

Due to the relatively reduced thickness of the edge region **250** in contrast to conventional approaches, RF power initially delivered to the ESC **215** has a high RF coupling

with the edge ring 206. In other words, the RF amplitude on the edge ring 206 could be higher than the RF amplitude on the substrate 203.

A gap 270 may be provided between an upper surface of the ceramic ring 250 and a lower surface of the edge ring 206. The gap 270 may be utilized to decrease coupling between the ring electrode and the plasma sheath 260 to reduce the RF current to edge tuning circuit 255. The thickness of the gap 270 may be selected to provide a desired amount of decoupling. In addition to the examples described above, other examples of the disclosure are also contemplated. In one example, the length of the transmission line 265 may have a length that is one half of the wavelength λ (i.e., $\lambda/2$) to facilitate matched impedance, in at least one frequency. In another example, the width of the edge control electrode 211 may be selected to increase or decrease electrical coupling with the edge ring 206, as desired. In another example, the gap 270 may be omitted. In another example, a conductive thermal gasket, for example, a silicone-based thermal gasket, may occupy the gap 270.

In another example, the edge tuning circuit 255 may be coupled to the match network 212 and then to the power source 213 instead of, or in addition to, ground. In such an example, the edge tuning circuit 255 would facilitate adjustment of capacitive coupling, rather than a parasitic effect as described above.

The edge tuning circuit 255 can include one or more variable capacitors as well as one or more inductors coupled to the edge control electrode 211. The resonant frequencies of the tuning circuit can be substantially close to the operating frequency, which enables a large variation of RF amplitude that is much larger and much smaller than the RF amplitude of the substrate.

FIGS. 3A-3B are cross-sectional views of an example substrate support assembly 300, in accordance with some embodiments. Substrate support assembly 300 can be used within a processing chamber to process a substrate (e.g., deposition, etching and/or lithography). As shown, substrate support assembly 300 includes a base structure including cooling plate 310 and hybrid puck 315 disposed on cooling plate 310. Hybrid puck 315 corresponds to an ESC.

In some embodiments, and as shown in FIG. 3A, hybrid puck 315 is bonded to cooling plate 310 via bonding layer 320. More specifically, hybrid puck 315 includes backing region 330 and chucking region 340, where backing region 330 is bonded to cooling plate 310 via bonding layer 320. Chucking region 340 is designed to securely hold a substrate. In some embodiments, chucking region 340 has a circular shape as viewed from the top of hybrid puck 315 to secure a circular substrate. In some embodiments, chucking region 340 has a rectangular shape as viewed from the top of hybrid puck 315 to secure a rectangular substrate. As shown, chucking region 340 can be formed to have a plateau shape having a center region that is elevated relative to a pair of end regions. In other words, the surface of the center region is elevated relative to the surfaces of the end regions.

In some embodiments, as shown in FIG. 3A, at least one bond protection structure 322 can be disposed between cooling plate 310 and backing region 330. For example, at least one bond protection structure 322 can include at least one O-ring and/or at least one gasket. At least one bond protection structure 322 can include any suitable material. In some embodiments, at least one bond protection structure 322 includes an elastomer. For example, at least one bond protection structure 322 can include a polyimide, a perfluoropolymer (PFP), etc.

Cooling plate 310 can have multiple cooling channels 312 embedded therein. Cooling channels 312 are pathways that allow a cooling fluid (e.g., water, fluorinert) to flow through hybrid puck 315 to dissipate heat during substrate (e.g., wafer) processing without interfering with the ability of hybrid puck 315 to hold the wafer securely in place. The purpose of cooling plate 310 is to keep the temperature of hybrid puck 315 and the substrate within a safe range to prevent damage to the ESC, the substrate and/or the rest of the processing chamber. The design and configuration of cooling channels 312 can depend on different variables, such as the structure of substrate support assembly and/or the manufacturing processes being used to process the substrate.

Backing region 330 is formed from a first dielectric material and chucking region 340 is formed from a second dielectric material different from the first dielectric material. For example, the first dielectric material can provide high thermal conductivity (which can enable improved heat flow to cooling plate 310) and the second dielectric material can offer leakage current stability. In some embodiments, the first dielectric material has a thermal conductivity that ranges between about 150 Watts per meter Kelvin (W/m-K) to about 200 W/m-K. For example, the first dielectric material can have a thermal conductivity of about 180 W/m-K. In some embodiments, the second dielectric material has a thermal conductivity that ranges between about 10 W/m-K to about 50 W/m-K. For example, the second dielectric material can have a thermal conductivity of about 30 W/m-K. In some embodiments, at least one of the first dielectric material or the second dielectric material is a ceramic material. For example, the first dielectric material can include AlN and the second dielectric material can include Al_2O_3 . In addition to enabling leakage current stability, the second dielectric material can enable improved scratch resistance, direct current (DC) and radio frequency (RF) separation, better RF coupling for low frequency, a wide operating temperature range, and/or a temperature-independent operation range.

In some embodiments, hybrid puck 315 has a thickness that ranges between about 5 millimeters (mm) to about 10 mm. In some embodiments, backing region 330 has a thickness that ranges from about 2.5 mm to about 5 mm. In some embodiments, chucking region 340 has a thickness that ranges from about 2.5 mm to about 5 mm. In some embodiments, the thickness of backing region 330 is approximately equal to the thickness of chucking region 340. In some embodiments, the thickness of backing region 330 is different from the thickness of chucking region 340.

In some embodiments, and as shown in FIG. 3A, backing region 330 is bonded to chucking region 340 via bond 345. For example, bond 345 can be a diffusion bond. As another example, bond 345 can be a metal bond. A metal bond may include an "interlayer" of aluminum foil which is placed in a bonding region between chucking region 340 and backing region 330. Pressure and heat may be applied to form a diffusion bond between the aluminum foil and chucking region 340 and between the aluminum foil and backing region 330. In another embodiment, a diffusion bond may be formed using other interlayer materials which are selected based upon the materials used for chucking plate and backing plate. In another embodiment, chucking region 340 may be directly bonded to backing region 340 using direct diffusion bonding in which no interlayer is used to form bond 345.

In some embodiments, as shown in FIG. 3B, at least one bond protection structure 347 can be disposed between backing region 330 and chucking region 340, and bond 345

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can be disposed between backing region **330** and chucking region **340**. For example, at least one bond protection structure **347** can include at least one O-ring and/or at least one gasket. At least one bond protection structure **347** can include any suitable material. In some embodiments, at least one bond protection structure **347** includes an elastomer. For example, at least one bond protection structure **347** can include a polyimide, PFP, etc.

As will be described in further detail below with reference to FIGS. **5A-5D**, a set of electrodes can be embedded within hybrid puck **315**. For example, chucking electrodes can be embedded within chucking region **340** to enable a substrate to be securely held to chucking region **340**. In some embodiments, an edge control electrode can be embedded within chucking region **340** to improve performance. For example, an edge control electrode can be a ring electrode. In some embodiments, a heating electrode is embedded within backing region **330**.

Substrate support assembly **300** can further include at least one band defining at least one cooling zone. In this example, the at least one band includes inner seal band **350-1** defining inner cooling zone **360-1** and outer seal band **350-2** defining outer cooling zone **360-2**. Each cooling zone can include cooling channels that allow for the cooling fluid to flow through substrate support assembly **300** for heat dissipation to prevent thermal stress on the substrate that can result in substrate defects (e.g., cracking or warping).

In some embodiments, instead of being bonded via bond **320**, hybrid plate **315** may be coupled to cooling plate **310** via one or more fasteners. In some embodiments, hybrid plate **315** includes multiple features. The features may match similar features in cooling plate **310** to which the hybrid plate **315** is mounted. Each feature accommodates a fastener. For example, a bolt (e.g., a stainless steel bolt, galvanized steel bolt, etc.) may be placed into each feature such that a head of the bolt is inside of an opening large enough to accommodate the head and a shaft of the bolt extends out of a bottom side of the hybrid plate **315**. The bolt may be tightened onto a nut that is placed in a corresponding feature in cooling plate **310**. Alternatively, features may be sized to accommodate a threaded insert such as a nut, and may include a hole that can receive a shaft of a bolt that is accommodated by a corresponding feature in the cooling plate. In another example, a helical insert (e.g., a Heli-Coil®) or other threaded insert (e.g., a press fit insert, a mold-in insert, a captive nut, etc.) may be inserted into one or more of the features to add a threaded hole thereto. A bolt placed inside of the cooling plate and protruding from cooling plate **310** may then be threaded into the threaded insert to secure cooling plate **310** to hybrid plate **315**. Alternatively, threaded inserts may be used in cooling plate **310**. In one embodiment, backing region **330** includes the features for accommodating threaded inserts and/or threaded fasteners, and chucking plate **340** lacks such features. Accordingly, the threaded inserts and/or heads of threaded fasteners may be fully encapsulated within hybrid plate **315**. The features may be slightly oversized as compared to a size of the fasteners to accommodate a greater coefficient of thermal expansion of the fasteners. In some embodiments, the fasteners are sized such that the fasteners will not exert a force on the features when the fasteners are heated to a temperature that ranges between about 250° C. to about 600° C.

Multiple sets of features may be included in hybrid plate **315**. Each set of features may be evenly spaced at a particular radius or distance from a center of hybrid plate **315**. For example, a first set of features may be located at a

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radius **R1** and a second set of features may be located at a radius **R2**. Additional sets of features may also be located at additional radii. In some embodiments, the features are arranged to create a uniform load on hybrid plate **315**. In some embodiments, the features are arranged such that a bolt is located approximately every 30-70 square centimeters (e.g., every 50 square centimeters).

The fasteners may be tightened to compress a bond protection structure (e.g., O-ring or gasket). The fasteners may all be tightened with approximately the same force to cause a separation between hybrid plate **315** and cooling plate **315** to be approximately the same (uniform) throughout the interface between hybrid plate **315** and cooling plate **310**. This may ensure that the heat transfer properties between hybrid plate **315** and cooling plate **310** are uniform.

FIG. **4** is a cross-sectional view of system **400**, in accordance with some embodiments. As shown, the system **400** includes substrate (e.g., wafer) **310** securely held to a surface of chucking region **340** of hybrid puck **315**, as described above with reference to FIG. **3**. More specifically, a voltage having a suitable magnitude can be applied to chucking electrodes embedded within chucking region **340** to generate a suitable electrostatic force to secure substrate **410** to chucking region **340**. Examples of hybrid pucks including sets of electrodes will now be described in further detail below with reference to FIGS. **5A-5D**.

FIG. **5A** is a cross-sectional view of hybrid puck **500A**, in accordance with some embodiments. As shown, hybrid puck **500A** includes backing region **510A** and chucking region **520A**, similar to backing region **330** and chucking region **340** described above with reference to FIGS. **3-4**. As further shown, a set of electrodes is embedded within chucking region **520A**. More specifically, the set of electrodes can include chucking electrodes **530A-1** and **530A-2** that can be used to generate an electrostatic force to securely hold a substrate (e.g., substrate **410** of FIG. **4**) disposed on chucking region **520A** upon application of a suitable voltage. As further shown, chucking electrodes **530A-1** and **530A-2** are electrically connected via connectors **532A-1** and **532A-2**. Accordingly, chucking electrodes **530A-1** and **530A-2** are dependent chucking electrodes.

The thickness of the portion of chucking region **520A** defined between the bottom surface of chucking region **520A** to chucking electrode **530A-1** is denoted by "A". The thickness of the portion of chucking region **520A** defined between chucking electrode **530A-1** and the surfaces of the end regions of the plateau shape of chucking region **520A** is denoted by "A'". The thickness of the portion of chucking region **520A** defined between chucking electrode **530A-2** and the surface of the center region of the plateau shape of chucking region **520A** (e.g., upper surface of chucking region **520A**) is denoted by "A''". In some embodiments, A, A' and A'' are each approximately equal. In some embodiments, at least one of A, A' or A'' is different.

FIG. **5B** is a cross-sectional view of hybrid puck **500B**, in accordance with some embodiments. As shown, hybrid puck **500B** includes backing region **510B** and chucking region **520B**, similar to backing region **330** and chucking region **340** described above with reference to FIGS. **3-4**. As further shown, a set of electrodes is embedded within chucking region **520B** and backing region **510B**. More specifically, the set of electrodes can include chucking electrodes **530B-1** and **530B-2** embedded within chucking region **520B** that can be used to generate an electrostatic force to securely hold a substrate (e.g., substrate **410** of FIG. **4**) disposed on chucking region **520B** upon application of a suitable voltage. In some embodiments, to improve performance, the set of

electrodes can include edge control electrode **535** embedded within chucking region **520B**. In some embodiments, edge control electrode **535** is a ring electrode, and can be independently controlled with respect to chucking electrodes **530B-1** and **530B-2**. Although not shown in FIG. 5A, chucking region **520A** can include a similar edge control electrode.

In some embodiments, the set of electrodes can further include heater electrode **540B** embedded within backing region **510B**. Heater electrode **540B** can include a material that can generate heat caused by the application of current from a power source (e.g., a metal material). Heat generated by heater electrode **540B** can maintain the temperature of the ESC and/or substrate. Heater electrode **540B** can work in conjunction with the cooling system (e.g., cooling channels **312** of cooling plate **310** of FIGS. 3-4) to control temperature throughout substrate processing. Although not shown in FIG. 5A, backing region **510A** can include a similar heater electrode. As further shown, chucking electrodes **530B-1** and **530B-2** are electrically connected via connectors **532B-1** and **532B-2**. Accordingly, chucking electrodes **530B-1** and **530B-2** are dependent chucking electrodes.

The thickness of the portion of chucking region **520B** defined between the bottom surface of chucking region **520B** to edge control electrode **535** is denoted by "B". The thickness of the portion of chucking region **520B** defined between edge control electrode **535** and the surfaces of the end regions of the plateau shape of chucking region **520B** is denoted by "B'". The thickness of the portion of chucking region **520B** defined between chucking electrode **530B-2** and the surface of the center region of the plateau shape of chucking region **520B** (e.g., upper surface) is denoted by "B'". In some embodiments, B, B' and B" are each approximately equal. In some embodiments, at least one of B, B' or B" is different.

FIG. 5C is a cross-sectional view of hybrid puck **500C**, in accordance with some embodiments. As shown, hybrid puck **500C** includes backing region **510C** and chucking region **520C**. In this example, backing region **510C** has a similar plateau shape as chucking region **520C**. More specifically, similar to chucking region **520C**, backing region **510C** can be formed to have a plateau shape having a center region that is elevated relative to a pair of end regions. In other words, the surface of the center region of backing region **510C** is elevated relative to the surfaces of the end regions of backing region **510C**.

As further shown, a set of electrodes is embedded within chucking region **520C** and backing region **510C**. More specifically, the set of electrodes can include chucking electrodes **530C-1** and **530C-2** embedded within chucking region **520C**. For example, chucking electrode **530C-1** can be used to power an edge ring (e.g., edge ring **114** of FIG. 1) and chucking electrode **530C-2** can be used to generate an electrostatic force to securely hold a substrate (e.g., substrate **410** of FIG. 4) disposed on chucking region **520C** upon application of a suitable voltage. Although not shown in FIG. 5C, chucking region **520C** can alternatively include an edge control electrode (e.g., ring electrode). In some embodiments, the set of electrodes can further include heater electrode **540C** embedded within backing region **510C**. As shown, heater electrode **540C** can follow the contour of the plateau shape of backing region **510C**. As further shown, chucking electrodes **530C-1** and **530C-2** are electrically connected via connectors **532C-1** and **532C-2**. Accordingly, chucking electrodes **530C-1** and **530C-2** are dependent chucking electrodes. The thickness of the portion of chuck-

ing region **520C** defined between the bottom surface of chucking region **520C** to chucking electrode **530C-1** is denoted by "C". The thickness of the portion of chucking region **520C** defined between chucking electrode **530C-1** and the surfaces of the end regions of the plateau shape of chucking region **520C** is denoted by "C'". The thickness of the portion of chucking region **520C** defined between chucking electrode **530C-2** and the surface of the center region of the plateau shape of chucking region **520C** (e.g., upper surface of chucking region **520C**) is denoted by "C'". In some embodiments, C, C' and C" are each approximately equal. In some embodiments, at least one of C, C' or C" is different.

FIG. 5D is a cross-sectional view of hybrid puck **500D**, in accordance with some embodiments. As shown, hybrid puck **500D** includes backing region **510D** and chucking region **520D**. In this example, backing region **510D** has a plateau shape. More specifically, backing region **510D** can be formed to have a plateau shape having a cross-sectional trapezoidal-shaped center region that is elevated relative to a pair of end regions. In other words, the surface of the center region of backing region **510D** is elevated relative to the surfaces of the end regions of backing region **510D**.

As further shown, a set of electrodes is embedded within chucking region **520D** and backing region **510D**. More specifically, the set of electrodes can include chucking electrodes **530D-1** and **530D-2** embedded within chucking region **520D** that can be used to generate an electrostatic force to securely hold a substrate (e.g., substrate **410** of FIG. 4) disposed on chucking region **520A** upon application of a suitable voltage. Although not shown in FIG. 5D, chucking region **520D** can include an edge control electrode (e.g., a ring electrode). In some embodiments, the set of electrodes can further include heater electrode **540D** embedded within backing region **510D**. As shown, heater electrode **540D** can follow the contour of the plateau shape of backing region **510D**.

As further shown, an opening is formed through hybrid puck **500D** corresponding to lift pin area **545** to accommodate a lift pin (e.g., 3 places). A lift pin is a component that can be used to place a substrate onto hybrid puck **500D** (e.g., from a robot apparatus) or raise a substrate for removal (e.g., by a robot apparatus). For example, the lift pin can be a metal pin that is mounted on a mechanism that can move the pin up or down (e.g., actuator or piston mechanism). Although not shown in FIGS. 5A-5C, at least one of hybrid puck **500A**, hybrid puck **500B** or hybrid puck **500C** can also include a similar lift pin area (e.g., 3 places). As further shown, hybrid puck **500D** can include a set of terminals including heater terminals **550-1** and **550-2** each connected to heater electrode **540D** and DC terminal **560** connected to chucking electrode **530D-2**. As further shown, chucking electrodes **530D-1** and **530D-2** are electrically connected via connectors **532D-1** and **532D-2**. Accordingly, chucking electrodes **530D-1** and **530D-2** are dependent chucking electrodes.

The thickness of the portion of chucking region **520D** defined between the bottom surface of chucking region **520D** to chucking electrode **530D-1** is denoted by "D". The thickness of the portion of chucking region **520D** defined between chucking electrode **530D-1** and the surfaces of the end regions of the plateau shape of chucking region **520D** is denoted by "D'". The thickness of the portion of chucking region **520D** defined between chucking electrode **530D-2** and the surface of the center region of the plateau shape of chucking region **520D** (e.g., the upper surface of chucking region **520D**) is denoted by "D'". In some embodiments, D,

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D' and D'' are each approximately equal. In some embodiments, at least one of D, D' or D'' is different.

FIG. 6 depicts a flow chart of an example method 600 for fabricating an ESC with a hybrid puck, in accordance with some embodiments. For example, method 600 can be performed to fabricate a portion of a substrate support assembly of a processing chamber to process a substrate.

At block 610, a base structure of an ESC is obtained. For example, the base structure can include a cooling plate having multiple cooling channels embedded therein. In some embodiments, obtaining the base structure includes forming the cooling plate including forming the multiple cooling channels.

At block 620, a hybrid puck is obtained. The hybrid puck can include a backing region and a chucking region disposed on the backing region. The backing region can be formed from a first dielectric material and the chucking region can be formed from a second dielectric material different from the first dielectric material. More specifically, the first dielectric material can be selected to improve thermal performance at the interface between the puck and the cooling plate. The second dielectric material can be selected to improve stability and leakage current consistency for the chucking surface. In some embodiments, at least one of the first dielectric material or the second dielectric material is a ceramic material. In some embodiments, first dielectric material includes AlN. In some embodiments, the second dielectric material includes Al₂O₃. In some embodiments, the chucking region and the backing region are bonded together with a bond. The chucking region and the backing region can be bonded together with any suitable type of bond. Examples of bonds include diffusion bonds, metal bonds, etc.

A set of electrodes can be embedded within the hybrid puck. For example, chucking electrodes can be embedded within the chucking region. In some embodiments, an edge control electrode is embedded within the chucking region. For example, the edge control electrode can be a ring electrode. In some embodiments, a heater electrode is embedded within the backing region. In some embodiments, forming the hybrid puck can include forming the set of electrodes within the hybrid puck. In some embodiments, at least a portion of the set of electrodes is embedded within the hybrid puck prior to forming the hybrid puck on the base structure (e.g., the chucking electrodes and/or edge control electrode within the chucking region, and/or the heater electrode within the backing region).

In some embodiments, the hybrid puck further includes a lift pin area (e.g., 3 places). For example, forming the hybrid puck on the base structure can include forming the lift pin area by forming a hole through the hybrid puck (e.g., from the elevated surface of the center region of the chucking region to the bottom surface of the backing region). In some embodiments, the hybrid puck further includes a set of terminals. For example, the set of terminals can include a heater terminal connected to a heater electrode and/or and a DC terminal connected to a chucking electrode.

The hybrid puck can further include at least one cooling zone disposed on the chucking region. In some embodiments, the at least one cooling zone includes an outer cooling zone defined by an outer seal band and an inner cooling zone defined by an inner seal band. For example, forming the hybrid puck on the base structure can include forming the at least one cooling zone on the chucking region.

In some embodiments, obtaining the hybrid puck includes obtaining a preassembled hybrid puck. In some embodi-

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ments, obtaining the hybrid puck includes forming the hybrid puck. For example, forming the hybrid puck can include embedding the set of electrodes within the backing region and the chucking region, forming the at least one cooling zone on the chucking region, and bonding the chucking region to the backing region using a bond.

At block 630, the hybrid puck is attached to the base structure. For example, the hybrid puck can be attached to the cooling plate. In some embodiments, attaching the hybrid puck to the base structure includes bonding the backing region of the hybrid puck to the base structure (e.g., cooling plate) via a bonding layer. In some embodiments, attaching the hybrid puck to the base structure includes coupling the backing region of the hybrid puck to the base structure using a set of fasteners. Further details regarding blocks 610-630 are described above with reference to FIGS. 1-5D.

The preceding description sets forth numerous specific details such as examples of specific systems, components, methods, and so forth, in order to provide a good understanding of several embodiments of the present invention. It will be apparent to one skilled in the art, however, that at least some embodiments of the present invention may be practiced without these specific details. In other instances, well-known components or methods are not described in detail or are presented in simple block diagram format in order to avoid unnecessarily obscuring the present invention. Thus, the specific details set forth are merely exemplary. Particular implementations may vary from these exemplary details and still be contemplated to be within the scope of the present invention.

Reference throughout this specification to "one embodiment" or "an embodiment" means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of the phrase "in one embodiment" or "in an embodiment" in various places throughout this specification are not necessarily all referring to the same embodiment. In addition, the term "or" is intended to mean an inclusive "or" rather than an exclusive "or." When the term "about" or "approximately" is used herein, this is intended to mean that the nominal value presented is precise within $\pm 25\%$.

Although the operations of the methods herein are shown and described in a particular order, the order of the operations of each method may be altered so that certain operations may be performed in an inverse order or so that certain operation may be performed, at least in part, concurrently with other operations. In another embodiment, instructions or sub-operations of distinct operations may be in an intermittent and/or alternating manner.

It is to be understood that the above description is intended to be illustrative, and not restrictive. Many other embodiments will be apparent to those of skill in the art upon reading and understanding the above description. The scope of the invention should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

What is claimed is:

1. A device comprising:

a hybrid puck corresponding to an electrostatic chuck, the hybrid puck comprising:

a backing region;

a chucking region disposed on the backing region, wherein the backing region comprises a first dielectric material to improve thermal performance of the hybrid puck, and wherein the chucking region com-

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prises a second dielectric material different from the first dielectric material to improve leakage current stability; and

a bond protection structure comprising an elastomer disposed between the backing region and the chucking region. 5

2. The device of claim 1, wherein at least one of the first dielectric material or the second dielectric material is a ceramic material.

3. The device of claim 1, wherein the first dielectric material is aluminum nitride, and wherein the second dielectric material is aluminum oxide. 10

4. The device of claim 1, wherein the chucking region is bonded to the backing region via a bond, and wherein the bond is one of a diffusion bond or a metal bond. 15

5. The device of claim 1, further comprising a set of electrodes embedded within the hybrid puck, wherein the set of electrodes comprises a chucking electrode embedded within the chucking region.

6. The device of claim 5, wherein the set of electrodes further comprises an edge control electrode embedded within the chucking region. 20

7. The device of claim 5, wherein the set of electrodes further comprises a heater electrode embedded within the backing region. 25

8. The device of claim 1, further comprising at least one seal band defining at least one cooling zone.

9. The device of claim 8, wherein the at least one seal band comprises an inner seal band defining an inner cooling zone and an outer seal band defining an outer cooling zone. 30

10. The device of claim 1, further comprising a base structure comprising a cooling plate having a plurality of cooling channels, wherein the hybrid puck is disposed on the base structure.

11. The device of claim 1, wherein the backing region and the chucking region each have a plateau shape defined by a respective center region elevated relative to a respective pair of end regions. 35

12. A substrate support assembly comprising:

a base structure comprising a cooling plate having a plurality of cooling channels; and 40

a hybrid puck corresponding to an electrostatic chuck and disposed on the base structure, the hybrid puck comprising a backing region disposed on the base structure, a chucking region disposed on the backing region, and a bond protection structure comprising an elastomer 45

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disposed between the backing region and the chucking region, wherein the backing region comprises a first dielectric material to improve thermal performance of the hybrid puck, and wherein the chucking region comprises a second dielectric material different from the first dielectric material to improve leakage current stability.

13. The substrate support assembly of claim 12, wherein the first dielectric material is aluminum nitride, and wherein the second dielectric material is aluminum oxide.

14. The substrate support assembly of claim 12, further comprising a bonding layer disposed between the base structure and the backing region.

15. The substrate support assembly of claim 12, wherein the chucking region is bonded to the backing region via a bond, and wherein the bond is one of a diffusion bond or a metal bond.

16. The substrate support assembly of claim 12, further comprising a set of electrodes embedded within the hybrid puck, wherein the set of electrodes comprises a chucking electrode embedded within the chucking region.

17. The substrate support assembly of claim 16, wherein the set of electrodes further comprises an edge control electrode embedded within the chucking region.

18. The substrate support assembly of claim 16, wherein the set of electrodes further comprises a heater electrode embedded within the backing region.

19. The substrate support assembly of claim 12, further comprising an inner seal band defining an inner cooling zone and an outer seal band defining an outer cooling zone.

20. A method comprising:

obtaining a base structure comprising a cooling plate comprising a plurality of cooling channels; and

attaching, to the base structure, a hybrid puck comprising a backing region disposed on the base structure, a chucking region disposed on the backing region, and a bond protection structure comprising an elastomer disposed between the backing region and the chucking region, wherein the backing region comprises a first dielectric material to improve thermal performance of the hybrid puck, and wherein the chucking region comprises a second dielectric material different from the first dielectric material to improve leakage current stability.

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